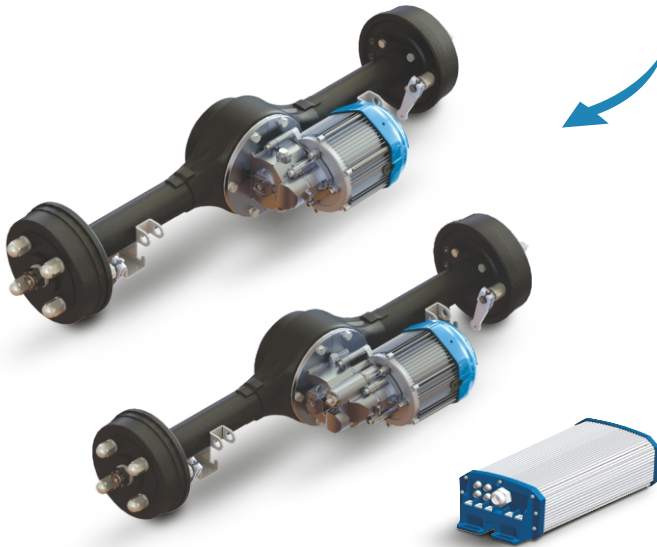
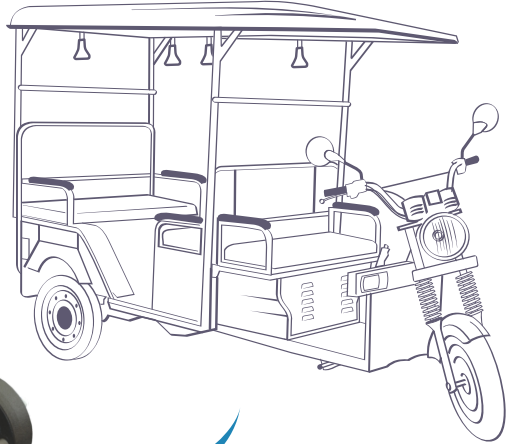


USER MANUAL

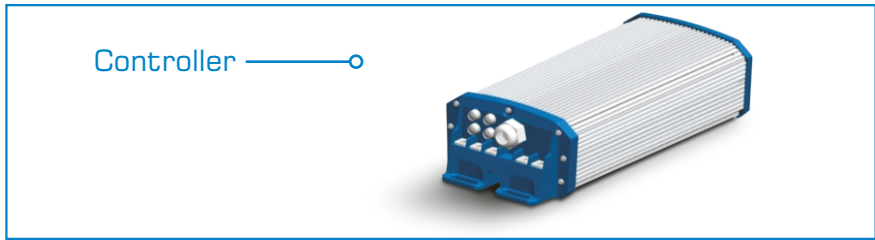
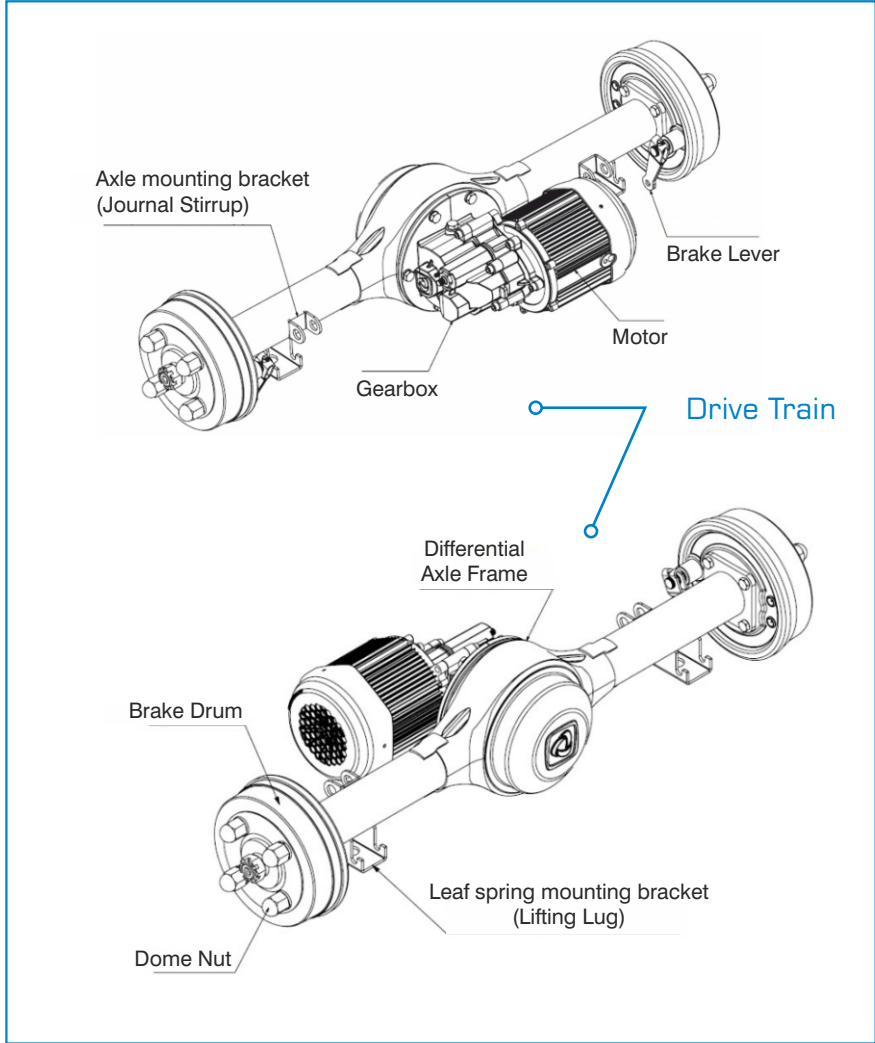
DRIVE TRAIN FOR e-RICKSHAW



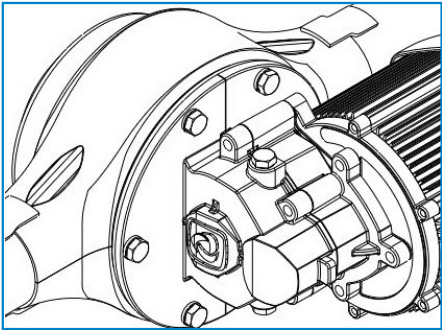
Model: BLDC – 1 Speed – 790/840/890

Model: BLDC – 2 Speed – 790/840/890

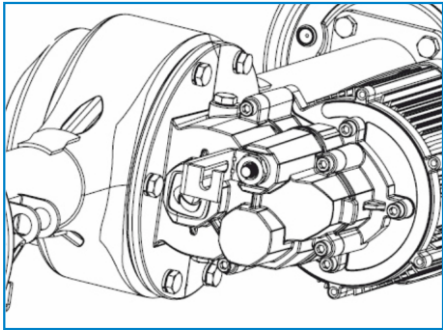
PRODUCT STRUCTURE



PRODUCT STRUCTURE

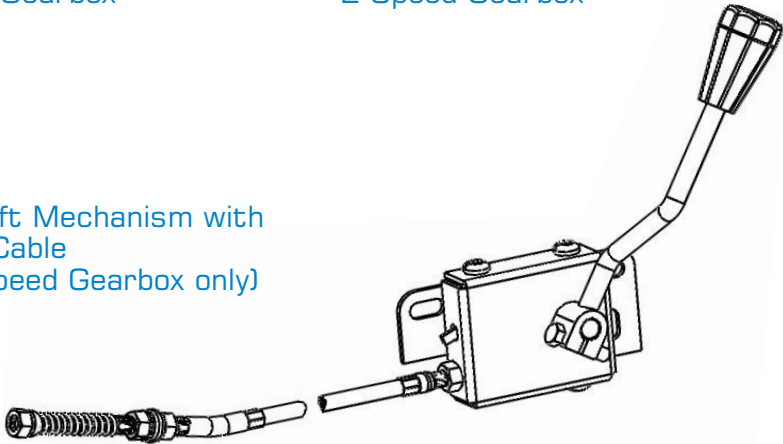


1 Speed Gearbox



2 Speed Gearbox

Gear Shift Mechanism with
Control Cable
(For 2 Speed Gearbox only)



Differential Axle Frame	
Parameter	Data
Length, mm	790 / 840 / 890
Brake Type	Drum Brake (Ø160mm)
Wheel Mounting, mm	4×M14 Stud @ 100 PCD
Chassis Mounting (Journal Stirrup)	Brackets provided
Leaf Spring Mounting (Lifting Lug)	Brackets provided

PRODUCT STRUCTURE

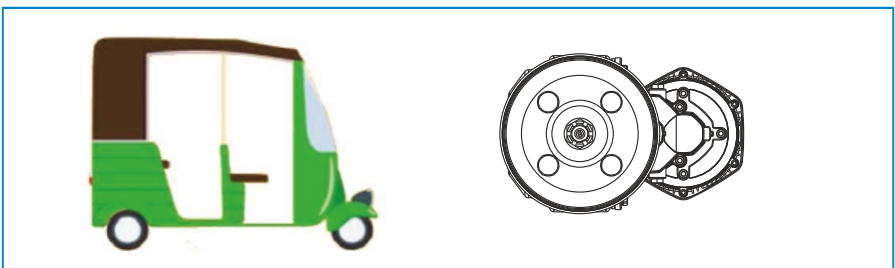
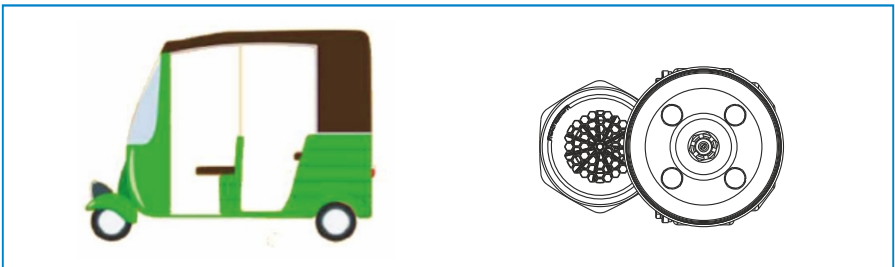
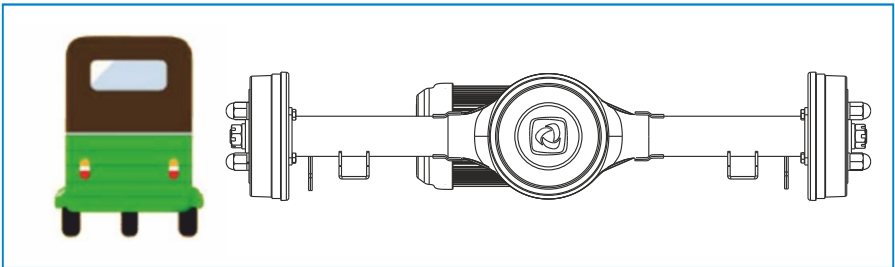
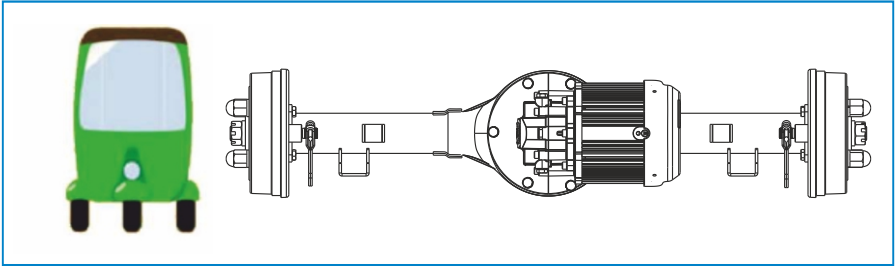
Controller Detail		
Parameter	1 kW	1.2 kW
Operating Voltage, VDC	48	
Max. Current, amp DC	55	60
No. of MOSFETs	24	
Low Voltage Protection, VDC	42 ± 1	
Ambient Temperature	-20° to +50° C	
Cooling Type	Natural Cooled	
Protection Class	IP54	
Efficiency	≥ 90%	

Motor Detail		
Parameter	1 kW	1.2 kW
Voltage, VDC	48	
Torque - Rated / Starting, Nm	3.5 / 21	4.1 / 24
Speed - Forward / Reverse, rpm	3000 / 1500	
Rated Current, amp DC	32	35
Peak Power, Watt	≤1900	≤ 2000
Rated Efficiency, %	≥ 78	≥ 80
Ingress Protection	IP54	

Gearbox Detail		
Closed configuration of Gearbox assures positive lubrication. Gearbox is supplied dully filled with lubricating oil EP 90.		
Parameter	1 Speed	2 Speed
Reduction Ratio / Vehicle Speed	10 : 1 / #25 kmph	Low Speed- 20.8 : 1 / #10 kmph High Speed- 8.3 : 1 / #25 kmph
Oil Capacity	150 ml	200 ml
Gear Shift Mechanism	Not Applicable	Required

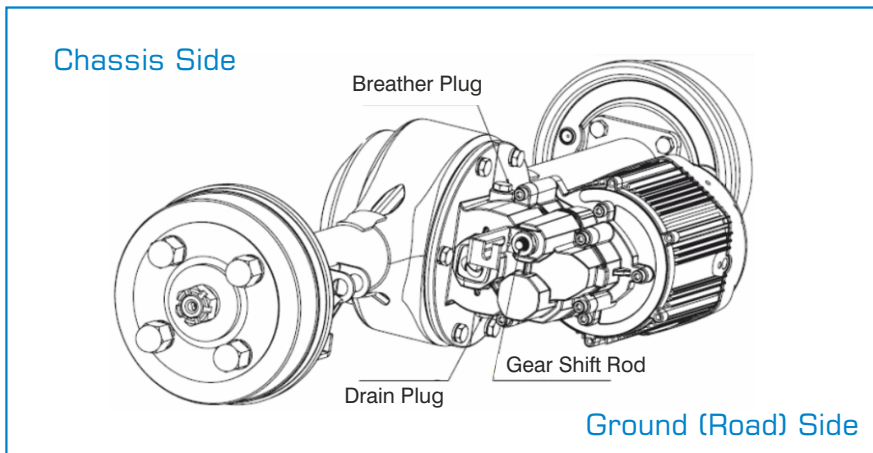
MOUNTING DETAILS

- Drive train is to be mounted on vehicle using Journal Stirrups and Lifting Lugs.
- Positioning of the drive train -

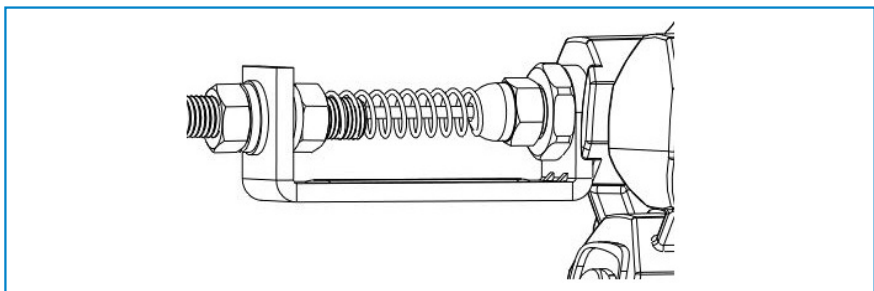


MOUNTING DETAILS

- Position of 2 Speed Gearbox



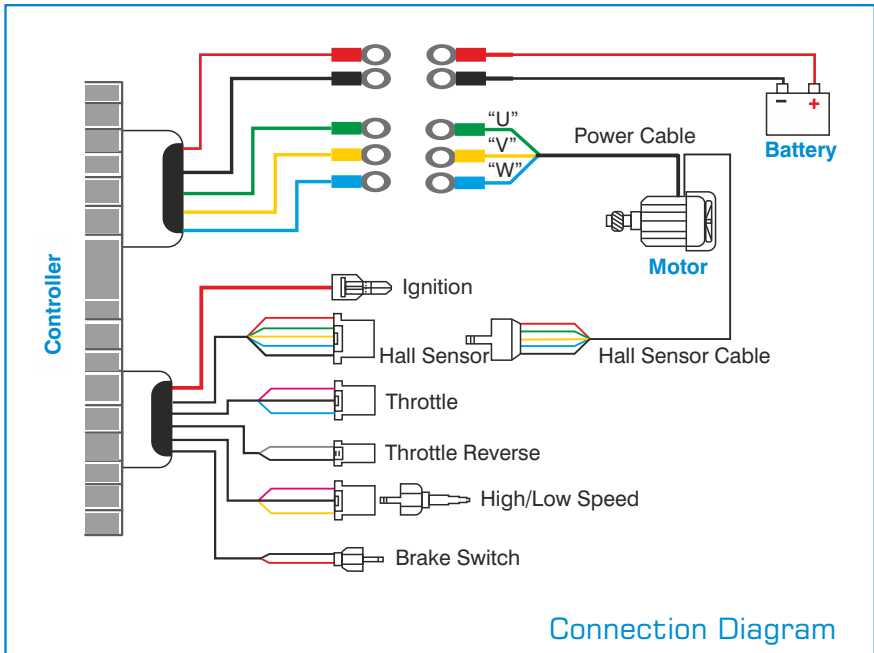
- In case of 2 Speed Gearbox, a Gear shift mechanism with lever is to be fixed in the driver's cabin.
- Gear shift mechanism is to be connected with Gear Shift Rod of gearbox using a control cable through Support plate.



- Paddle brake is to be attached to the brake levers with the connecting rods.
- Controller is to be mounted in a well ventilated space by fixing nut bolt in the end cover.

CONNECTION DETAILS

- Battery is to be connected to the controller using a junction box through a protective device.
- Motor is to be connected to the controller using a junction box.
- All other functions are to be connected as per diagram.

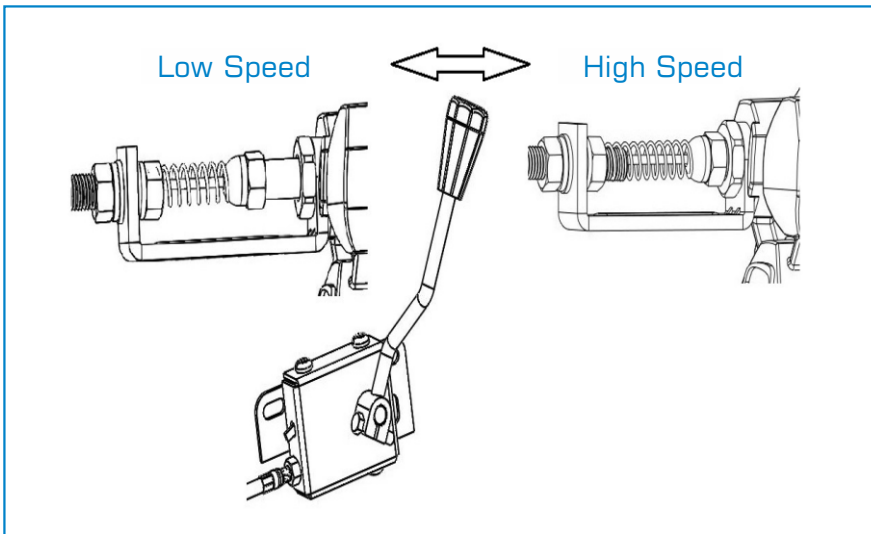


OPERATING INSTRUCTIONS

- Turn ON main DC power to Controller.
- Switch ON ignition.
- While switching F/R switch of throttle in forward mode, the vehicle will run forward at rated speed. While switching the F/R switch in reverse mode, the vehicle will run backwards at 50% of the rated speed.
- In High/Low speed function, a link is provided for shorting the terminals. Different connections of connector, result in different speeds as below:
 - # Connector unplugged – Medium speed (80%)
 - # Connector violet & black terminals shorted – Full Speed (100%)
 - # Connector white & black terminals shorted – Low Speed (60%)

OPERATING INSTRUCTIONS

- Two terminals of brake switch connector are to be connected in series with the mechanically operated Brake Switch. While vehicle brake is applied, the mechanically operated Brake Switch gets operated which, in turn, cuts off the supply to motor from controller. This is meant for motor safety while brake is applied.
- In case of Gear Train with 2 Speed Gearbox, high & low speed can be achieved by actuating lever of Gear Shift Mechanism



DOs & DON'Ts

- Check condition of breather plug periodically to ensure that it is not clogged.
- Check condition of Brake liner after 1000 km, 10000 km & 20000 km for both safety & performance point of view.
- Avoid heavy braking (high acceleration followed by sudden braking) to avoid faster wear of Brake liners.
- Modification made to components of system or replacement of original components can affect the performance.
- Avoid overloading of vehicle to maintain condition of components and consistent performance of Drive Train.



e - RICKSHOW
FOR POLLUTION FREE
& ECONOMIC
TRANSPORTATION



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ROTODRIVE is a brand of Rotomotive for e-mobility products.